Case Study 80: Public Transport Pass Management System

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Problem Statement

Current manual transport pass management systems create operational bottlenecks and frustrate both administrators and passengers. The lack of automated tracking makes it nearly impossible to efficiently monitor pass validity, leading to revenue loss and service disruptions.

Inefficiency

Manual data entry and record-keeping consume excessive time and resources

Tracking Difficulties

Locating and verifying pass validity requires searching through physical files

Slow Renewals

Passengers face long wait times and delays when renewing their transport passes

Error-Prone

Human errors in manual entry lead to incorrect records and billing issues

Solution: Transport Pass Management System (C++)

Key Features

- Apply for new transport passes by entering passenger details
- Validate pass authenticity using a unique **Pass ID**
- Renew existing passes and update validity and price
- Automatically apply **student discount** when applicable
- Menu-driven interface for easy navigation

Stored Data Elements

Pass ID, Holder Name, Age, Pass Type, Pass Plan, Validity Days, and Calculated Price

System Architecture & Working Method

1 Display Menu

The system presents available options: Apply for Pass, Check Pass, Renew Pass, or Exit.

2 User Selects Option

The user chooses a desired action from the menu using input.

3 Apply Pass

For new passes, the user enters passenger details (name, age, pass type, etc.) to create a new record.

4 Check Validity

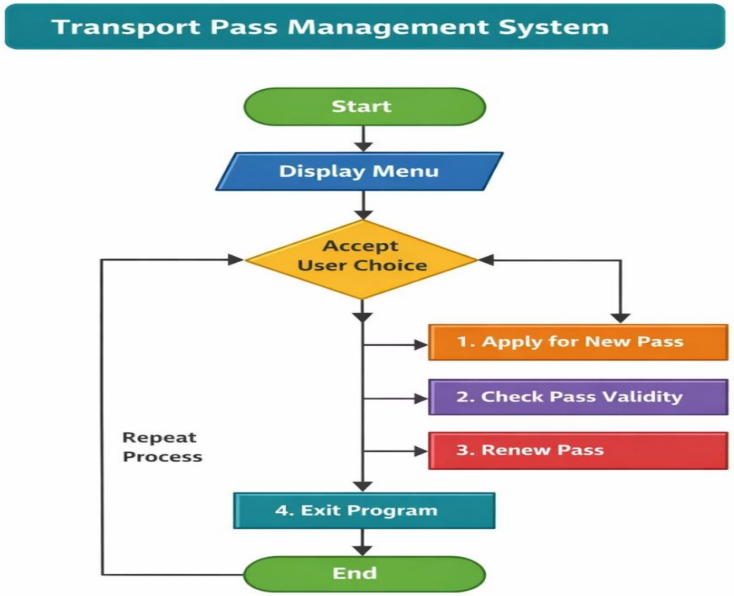
The system allows searching and retrieving pass information using a unique Pass ID.

5 Renew Pass

For existing passes, the user can update the validity period, and the price is recalculated.

6 Program Repeats Until Exit

After each operation (Apply, Check, Renew), the system returns to the main menu. The program terminates when the 'Exit' option is selected.



Conclusion & Next Steps

Current Status

Functional console-based system demonstrating core transport pass management capabilities with complete data handling

Immediate Goals

Implement file persistence to maintain data across program sessions and expand validation features

Long-term Vision

Develop comprehensive transport ecosystem with mobile integration, payment gateway, and fleet management

This C++ Transport Pass Management System successfully addresses manual management challenges while providing a scalable framework for future enhancements.

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Thank You